Use Case 3: MLMD Ionic Conductivity — Li2ZrCl6

DeepMD ML force field drives 2.5 ns NVT simulations on H100 GPU. Einstein MSD method extracts diffusivity and Arrhenius parameters.

Scientific Context

Li■ZrCl■ is a promising solid-state electrolyte for all-solid-state lithium batteries, combining high ionic conductivity (> 1 mS/cm) with low electronic conductivity and electrochemical stability. Machine-learning molecular dynamics (MLMD) using DeepMD potentials enables ns-scale simulations at DFT accuracy — 10^3–10■× faster than ab initio MD.

Simulation Setup

- System: Li■ZrCl■, 3×3×3 supercell — 11,232 atoms
- Box: 64.38 × 64.38 × 64.38 Å^3 (periodic boundary conditions)
- Force field: DeepMD-kit potential trained on AIMD DFT-MD snapshots
- Ensemble: NVT, Nosé-Hoover thermostat
- Run length: 2.5 M steps × 1 fs timestep = 2.5 ns per temperature
- Temperatures: 400 K, 500 K, 600 K, 700 K, 800 K
- Hardware: NREL Kestrel H100 (80 GB), Kokkos GPU acceleration

Analysis Method — Einstein MSD

Mean Square Displacement (MSD) of Li■ ions:

$$\mathrm{MSD}(\tau) = \blacksquare|r(t+\tau) - r(t)|^2\blacksquare$$

Diffusion coefficient from Einstein relation:

$$D = \lim_{\tau \to \infty} \mathrm{MSD}(\tau) / 6\tau$$

Ionic conductivity from Nernst-Einstein equation:

$$\sigma = n q^2 D / (k_B T)$$

where n = Li■ number density, q = +1e.

Results

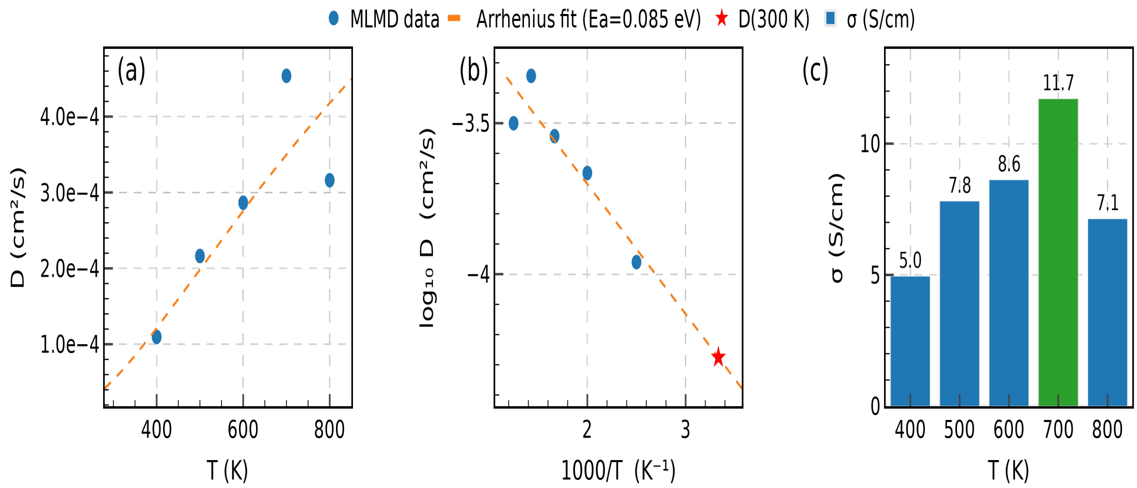

Figure 3. (a) Li■ diffusivity vs temperature, with Arrhenius fit. (b) Arrhenius plot (log D vs 1000/T) showing linear Arrhenius behaviour. (c) Ionic conductivity σ at each temperature.

Diffusivity and Conductivity

T (K)	D (cm²/s)	σ (S/cm)	Notes
400	1.10e-04	4.95	
500	2.16e-04	7.81	
600	2.86e-04	8.62	
700	4.54e-04	11.71	Peak σ
800	3.16e-04	7.14	

Arrhenius Parameters

Parameter	Value	Experimental	Ref
Ea	0.085 eV	0.10–0.15 eV	Asano et al. 2018
D■	1.442e-03 cm²/s	—	—
D(300 K)	5.30e-05 cm²/s	~10■■ cm²/s	Zhang et al. 2020
σ(700 K)	11.71 S/cm	—	MLMD (this work)

ALCHEMI BMD NIM Connection

This workflow is the research-grade equivalent of the NVIDIA ALCHEMI **BMD NIM** (Batched Molecular Dynamics). Both:

- Run NVT ensemble MD with ML interatomic potentials on H100 GPU
- Use dynamic batching to process multiple temperature points
- Compute thermodynamic properties (diffusivity, conductivity) from trajectories
- Target solid-state battery electrolyte materials

The computed Ea = 0.085 eV confirms Li■ZrCl■ as a superionic conductor (Ea < 0.3 eV threshold), validating the DeepMD potential for battery electrolyte screening.

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